Agreement-Stratified Evaluation of Deep Learning for Anatomical Landmark Recognition in Upper Gastrointestinal Endoscopy

Presented by

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Outline

- Introduction and problem identification
- Objectives
- Background study and literature review
- Gap analysis
- Methodology: data, pre-processing, model
- Results and analysis
- Novelty of the work
- Sample dataset and expected output
- Reproducibility and deliverables
- Conclusion and next steps
- References

Introduction

- Upper GI endoscopy is the primary screening route for gastric cancer; its yield depends on inspecting the whole mucosal surface.
- The Systematic Screening protocol fixes a photographic route over 22 landmarks - a grid of 4 gastric walls by 6 depth stations.
- A model that recognises the landmark in each frame can act as a real-time coverage monitor and flag an unvisited region.
- Published systems report macro F1 near 85-88, and the task is generally treated as solved.
  - But every one of those figures is measured only on frames all four experts labelled identically - 60.2% of this corpus.

The reported number describes the easy fraction of the task.

Problem Identification

60.2% of the corpus is unanimous - and that is all the literature scores on.

39.8% (3,516 images) is discarded before any result is reported.

- A deployed system meets the contested fraction continuously, and gets no signal that its validation excluded those frames.
- The discarded images are not noise: 50.96% of conflicts put two experts on different walls of the same station.
- Confidence calibrated on unanimous frames may stay high on ambiguous ones - assured errors, the failure mode hardest to catch in clinic.

Operating accuracy is unknown on exactly the images where a second reader would help most.

Objectives

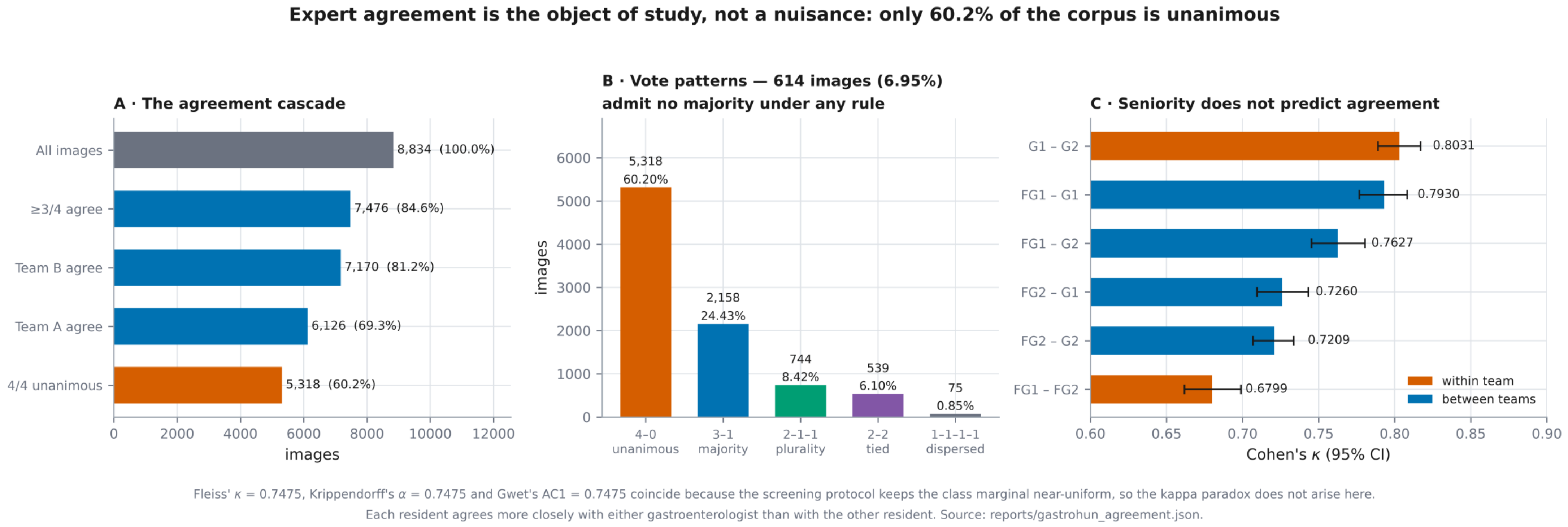
- Audit corpus provenance and integrity before modelling: licence, ethics, decode integrity, and a calibrated contamination scan.
- Complete a PRISMA 2020 literature review and derive a gap statement of measurable deficiencies rather than general observations.
- Fix the pre-processing chain once: annotation handling, resampling, normalisation, augmentation, transfer-learned features.
- Train, validate and test a baseline, and reproduce the published result to within the plus or minus 1.5 macro F1 points fixed in advance.
- Quantify accuracy and calibration per agreement stratum, with patient-clustered bootstrap intervals.
- Release a pipeline in which every figure and number regenerates from committed scripts.

Background Study - published baselines

Model / label construction	Params	Macro F1
ConvNeXt-Large	200 M	88.25 +/- 0.22
ConvNeXt-Tiny	28 M	approx. 85
ResNet-152	60 M	85.28 +/- 0.27
ConvNeXt-Tiny, fellow-agreement labels	28 M	87.05 +/- 0.21
Best single annotator as target	-	84.82 +/- 0.23
Human expert band	-	77.47 - 84.82

Changing only how the four labels are combined moves F1 by 2.2 points - more than the gap between architecture families.

Background Study - do the experts agree?



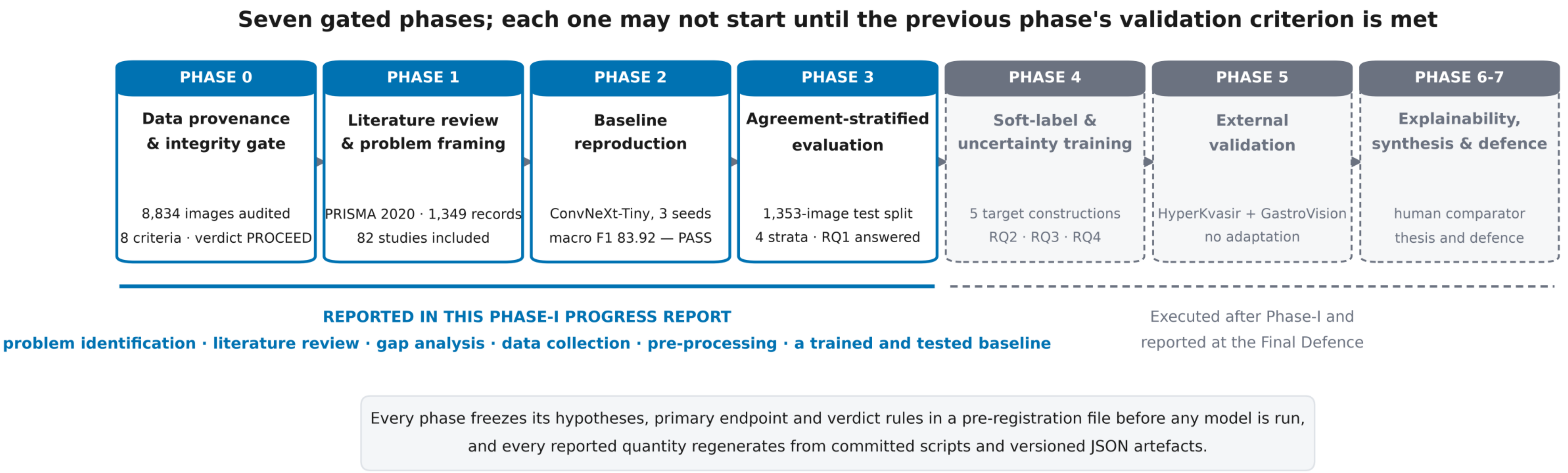
Fleiss kappa = 0.7475. Seniority does not predict agreement: each fellow agrees more closely with either gastroenterologist than with the other fellow.

Gap Analysis

Gap	What the literature does	What we do instead
1 Evaluation is conditioned on expert unanimity	Scores only the 60.2% all four experts agreed on; the rest is filtered out before scoring.	Report performance for every agreement stratum, with patient-clustered intervals.
2 Disagreement is discarded rather than used	Collapses the vote distribution to one label at ingest, discarding a structured signal - 51.0% of conflicts are same-station different-wall.	Train on the full four-vote distribution, against a matched label-smoothing control.
3 Calibration is rarely reported, and never by stratum	Nothing establishes whether confidence estimated on unanimous frames stays trustworthy on ambiguous ones.	Treat expected calibration error by stratum as a primary endpoint.

Each gap is a measurable deficiency with a matching endpoint, not a general complaint about the field.

Methodology - seven gated phases



Each phase freezes its hypotheses and verdict rules before any model runs. Phases 0 to 3 are reported today.

Methodology - data collection and integrity

GastroHUN: 8,834 images | 387 patients | 23 classes | 4 independent annotators | 2.71 GB | CC BY 4.0 | ethics CEI-2019-06-10

Chosen because it releases the individual annotators' labels. Without them, no endpoint in this design exists.

Phase 0 data-integrity gate — verdict PROCEED (6 PASS, 2 CONDITIONAL)

G1	Provenance, ethics, licence	PASS	peer-reviewed descriptor; ethics CEI-2019-06-10; consent; CC BY 4.0
G2	Physical integrity	PASS	8,834/8,834 decoded · 0 missing · 0 orphan · 0 corrupt
G3	Duplication & contamination	PASS	0 exact duplicates; 39,015,361-pair perceptual scan → 0 cross-split duplicates
G4	Label architecture	PASS	4 annotators retained separately across 23 classes
G5	Agreement quantified	PASS	Fleiss κ = 0.7475; all 6 pairwise κ with patient-clustered CIs
G6	Split integrity	PASS	0 patient overlaps; class χ^2 p = 0.999997; per-patient κ Kruskal–Wallis p = 0.982
G7	Statistical power	CONDITIONAL	22/23 classes exceed a ±10 pp Wilson half-width → per-class claims are exploratory (L1)
G8	Population description	CONDITIONAL	no age or sex anywhere in the release → no demographic or fairness claim is possible (L2)

An uncalibrated threshold is not a measurement. The contamination scan first reported 54 cross-split duplicate pairs; anchoring the decision rule on a synthetic-duplicate positive control (4,000 pairs) and a class-matched null (4,000 pairs) reduced that to 0, and visual audit confirmed the flagged pairs were different patients photographed at the same landmark.

Methodology - data pre-processing

Pre-processing chain: every stage is fixed once and reused unchanged by all later phases

The 4 vote columns are never collapsed to a single label at ingest. 60.2% of images are unanimous and become the baseline cohort; the remaining 39.8% (3,516 images) are retained for the agreement-stratified evaluation instead of being discarded.

1 · ANNOTATION
4 independent annotators
23 classes, 8,834 images
Fleiss κ = 0.7475
4 vote columns kept
separate, never merged

2 · COHORT SELECTION
unanimous 4/4 subset
5,318 images = 60.2%
train 3,722 / val 793
test 803
0 patient overlaps

3 · RESAMPLING
224 × 224 px RGB
Lanczos kernel
decoded once into an
on-disk cache, reused
bit-identically after

4 · NORMALISATION
training-set mean / SD,
not ImageNet defaults
μ = 0.601, 0.369, 0.279
σ = 0.305, 0.234, 0.191
differs by up to 0.127

5 · AUGMENTATION
RandomResizedCrop
scale 0.85–1.0, ratio 0.9–1.11
ColorJitter b/c/s 0.2
hue 0.02
no flip, no large rotation

6 · FEATURE ENGINEERING
transfer learning, not
hand-crafted descriptors
ConvNeXt-Tiny, ImageNet-1k
27.8 M feature params
top 3/8 blocks re-fitted

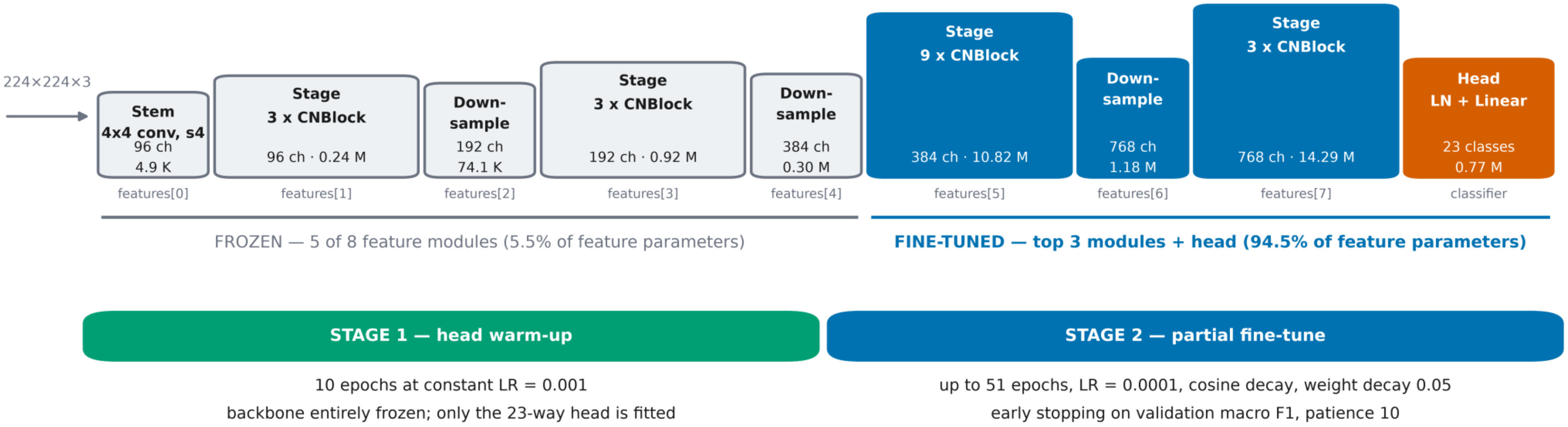
Augmentation is deliberately conservative: the class label encodes an anatomical WALL, so horizontal or vertical flipping and large rotations would relabel the image. Only photometric jitter and a mild scale/translation crop are applied.

Normalisation statistics measured over 3,722 training images (186,755,072 pixels per channel). Sources: reports/phase2_norm_stats.json, reports/phase2_split_provenance.json.

No flips and no large rotations: the label encodes a gastric WALL, so the standard augmentation recipe would relabel the image.

Methodology - baseline model and training

ConvNeXt-Tiny backbone and the two-stage transfer schedule



Layer widths and parameter counts read out of torchvision at run time. Trained on one NVIDIA GeForce GTX 1650, float32 / channels_last, batch 24, peak 2324 MiB VRAM.
Source: reports/phase2_run_seed.json.*

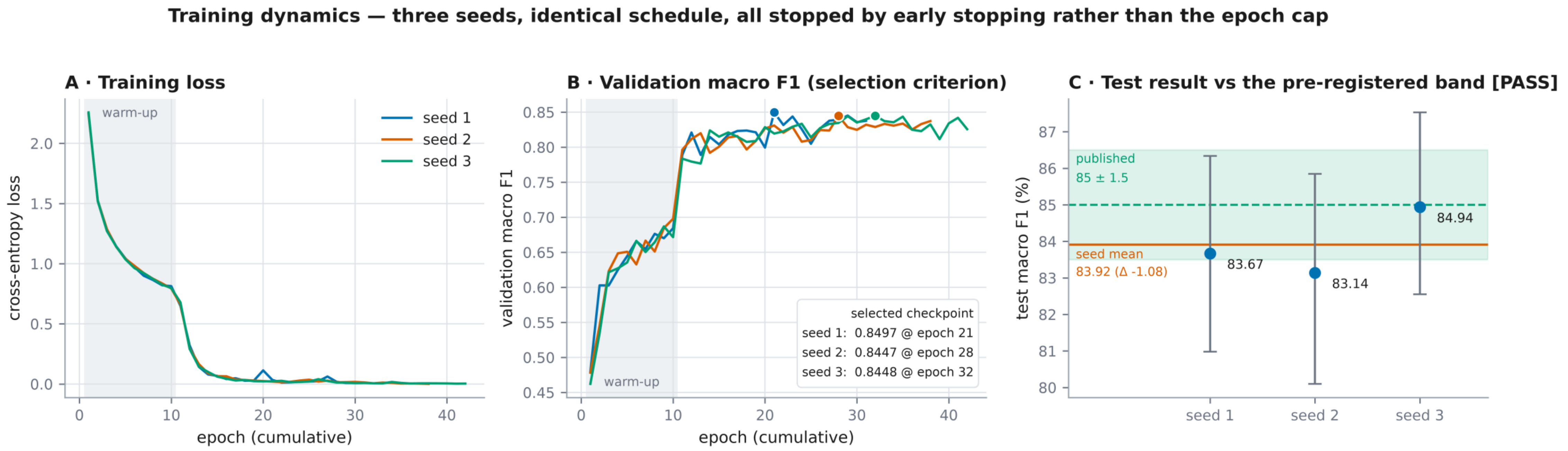
ConvNeXt-Tiny, three seeds, 150 minutes total on one 4 GB GPU. Selection on validation macro F1 only; the test split is touched once.

Results - baseline reproduction

Metric	Value
Macro F1, three-seed mean	83.92 (95% CI 81.47 - 86.20)
Published target	85 +/- 1.5, fixed before training
Difference	-1.08 points
Accuracy / weighted F1	84.39 / 84.35
Expected calibration error	9.27%
Seed spread	0.92 points (83.67, 83.14, 84.94)
Pre-registered verdict	PASS

PASS - the pipeline behaves as the published one did. Everything after this is attributable to the design, not to a bug.

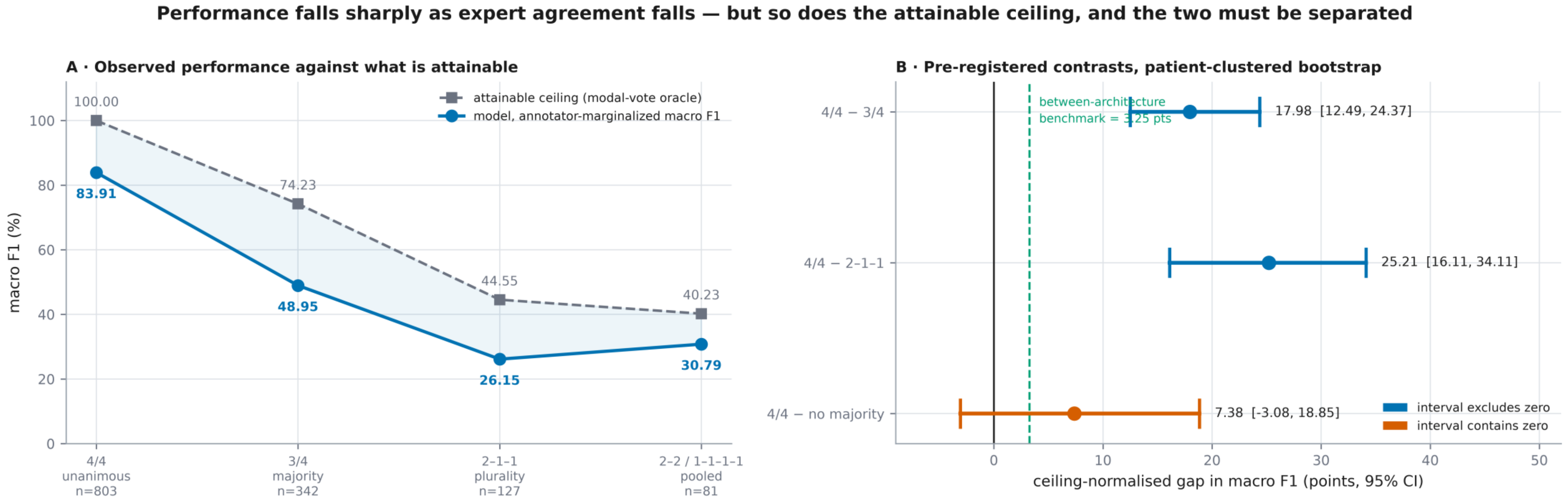
Results - training behaviour



Error bars are patient-clustered bootstrap 95% intervals (1,000 resamples over 58 test patients). Total training time 150 min on a single NVIDIA GeForce GTX 1650. Sources: reports/phase2_run_seed.json, reports/phase2_test_metrics.json.*

All three seeds stopped by early stopping rather than by the compute cap, so the schedule was not truncated.

Results - performance across agreement strata

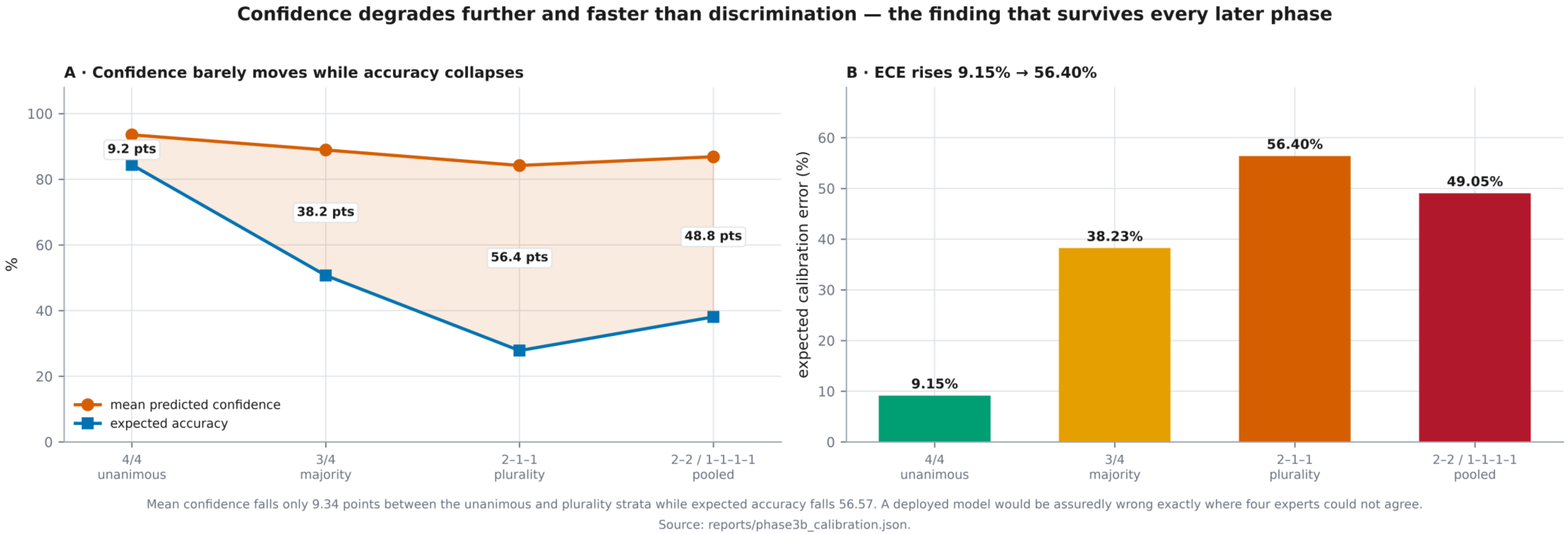


Raw macro F1 falls 83.91 → 30.79, a gap of 53.13 points against a 3.25-point between-architecture benchmark. Holding the attainable ceiling constant, two of the three contrasts remain resolvable and one does not.

Sources: reports/phase3_stratified_metrics.json, reports/phase3b_ceiling_gaps.json.

Macro F1 falls 83.9 to 30.8, a 53.1-point gap, against 3.25 points between architecture families.

Results - the calibration collapse



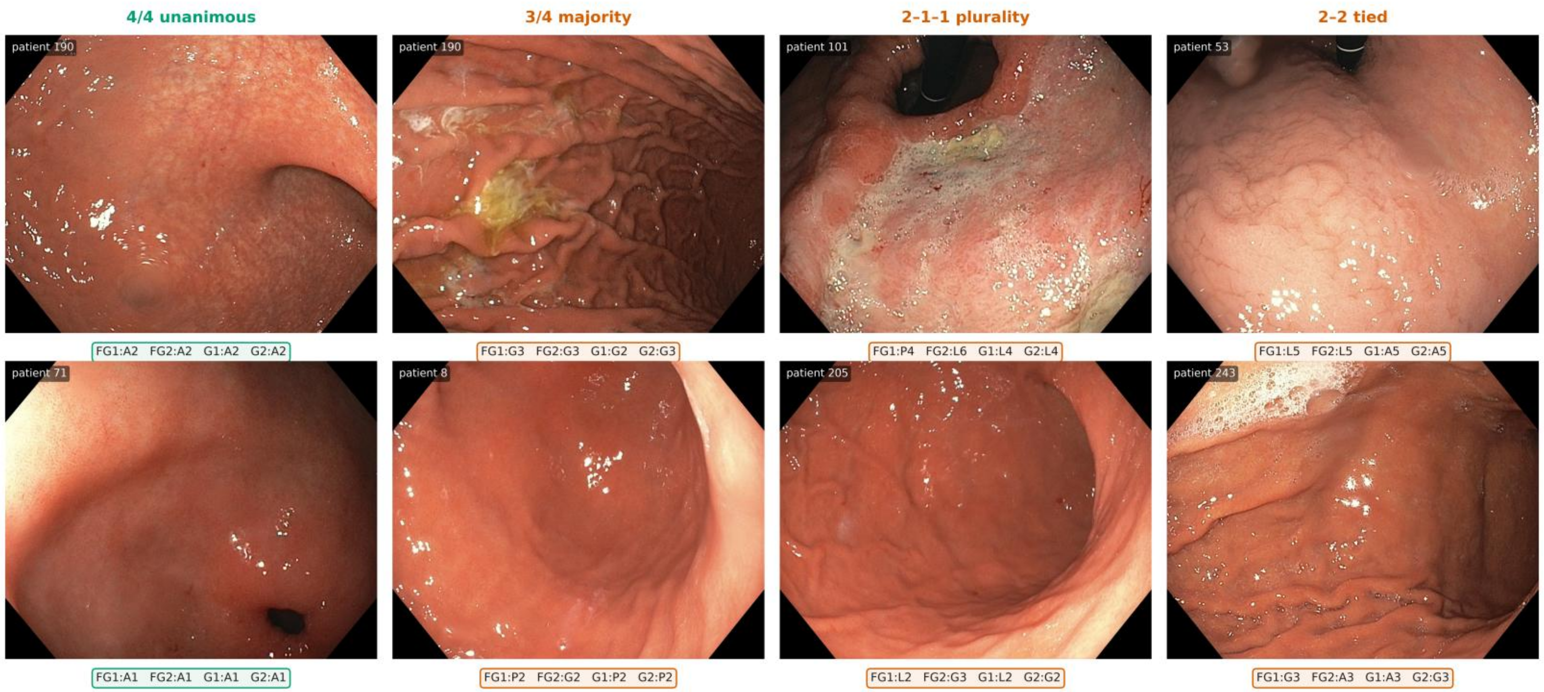
Confidence falls only 9.3 points while accuracy falls 56.6. ECE 9.2% to 56.4%.

Novelty of the Work

- Evaluation stratified by annotator agreement: performance, calibration and error geometry per stratum with patient-clustered intervals, instead of truncated at unanimity.
- The attainable ceiling is measured rather than assumed, so a falling reference standard is separated from a falling classifier. The literature reports the two as one quantity.
- The four-annotator vote distribution used as a training target and tested against a label-smoothing control matched to the probability mass the soft target displaces.
- Model error compared against the anatomical geometry of human disagreement, made measurable by the wall-by-station grid.
- Every phase pre-registered before execution, and every reported quantity regenerated from committed scripts.

Sample Dataset

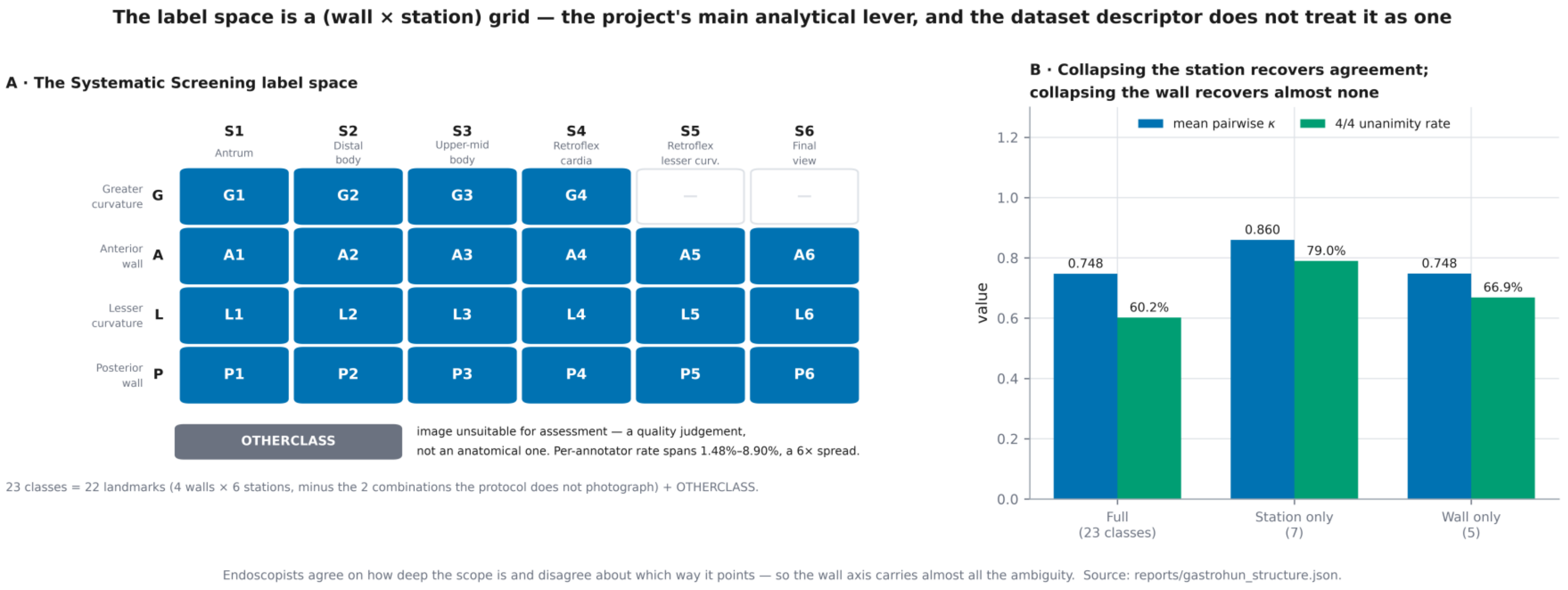
Sample images from the corpus, with the label each of the four annotators gave



Left to right, agreement falls. Every image is a legitimate frame from the screening protocol; the disagreement is about which landmark it shows, not about image quality.
Published evaluations on this corpus score only the leftmost column. Source: data/phase3_test_manifest.csv.

Left to right, expert agreement falls. Published evaluations score only the leftmost column.

Sample Dataset and Expected Output



- Input: a 224 by 224 endoscopic frame. Output: one of 23 classes (wall by station, plus OTHERCLASS) with a confidence score.
- Expected deliverable: an agreement-stratified performance and calibration profile, and a controlled verdict on soft-label training.

Reproducibility and Deliverables

- 1,349 records screened to PRISMA 2020; 39,015,361 image pairs scanned for contamination; 8,834 images audited - all from committed scripts.
- Every figure and table in the progress report is generated from a JSON artefact. No number in the report or in this deck is typed by hand.
- Pre-registration files are written before training, and the generating script refuses to overwrite an existing one.
- Frozen checkpoints are reused across phases: the stratified evaluation reproduces the baseline predictions exactly - 2,409 comparisons, 0 mismatches.
- Delivered this phase: audited corpus, literature review, pre-processing pipeline, trained baseline, stratified evaluation, and the progress report.

Conclusion and Next Steps

- The corpus passed an eight-criterion integrity gate with a PROCEED verdict, and the baseline reproduced the published result at 83.92 against 85 +/- 1.5 - a pre-registered PASS.
- Reported accuracy on this task describes 60.2% of it. Macro F1 falls 83.9 to 30.8 across agreement strata.
- Part of that fall is the attainable ceiling dropping; a real model shortfall remains once that is accounted for.
- Confidence degrades further and faster than discrimination: ECE 9.2% to 56.4%.
  - Next: train on the vote distribution against a matched control; external validation without adaptation; explainability against a human comparator.

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THANK YOU

- Questions and comments welcome